MarsLab: A Martian Rover Simulator for Planetary Rover Autonomous Navigation

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Abstract— **Future Mars missions will require rover autonomy that can operate across unstructured terrain, changing illumination, atmospheric dust, and limited communication. Simulation is a practical way to study these conditions before deployment, but existing Mars-relevant resources differ in scope, including mission-oriented simulators, fixed analog datasets, task-specific environments, and open robotics interfaces. In this context, we present MarsLab, an open-source, ROS2-native Mars rover simulator for autonomy and navigation algorithm development. MarsLab combines HiRISE-derived and procedural terrain with customizable rock, crater, solar-illumination, and atmospheric-dust settings, and runs a Perseverance-class rover model in NVIDIA Isaac Sim. The runtime publishes RGB, depth, RGB-D point clouds, LiDAR, IMU, wheel odometry, and Ground Truth (GT) pose data through standard ROS2 topics. We demonstrate MarsLab with Simultaneous Localization and Mapping (SLAM) benchmarks across sensing modalities, dust levels, scene geometry, and route length, and with Visual Place Recognition (VPR) benchmarks over repeated Mars Base traversals under illumination and dust changes. The results illustrate how controlled scene variation and shared GT trajectories can be used to compare trajectory-level estimation and image-level place recognition within the same simulator. Our Project Page: `https://kimhoyun-robotair.github.io/MarsLab.github.io/`.**

I. Introduction

Mars exploration is entering a renewed robotic phase. In 2026, NASA approved implementation of its support to ESA's Rosalind Franklin rover and selected SpaceX Falcon Heavy for the mission's late-2028 launch opportunity [1]. NASA also announced Space Reactor-1 Freedom, a nuclear-electric spacecraft intended to reach Mars before the end of 2028 and deploy a Skyfall payload of Ingenuity-class helicopters [2]. Together with Perseverance AutoNav and Ingenuity, these plans point toward Mars missions that combine rover mobility, aerial scouting, commercial launch, and increasingly autonomous robotic decision making [3], [4].

Mars exploration remains difficult despite this momentum. Mission systems have progressed from Mars Exploration Rover visual odometry [5] to Perseverance driving autonomy [3], orbital-map localization [6], and coupled SLAM/navigation pipelines [7]. Yet surface autonomy must operate without Global Navigation Satellite System (GNSS), with delayed communication, and over low-texture regolith, repetitive terrain, changing illumination, dust, and tight resource budgets [8]. Localization is especially limiting: pose error propagates into mapping and planning, while repeatable ground-truth pose data on Mars remains scarce.

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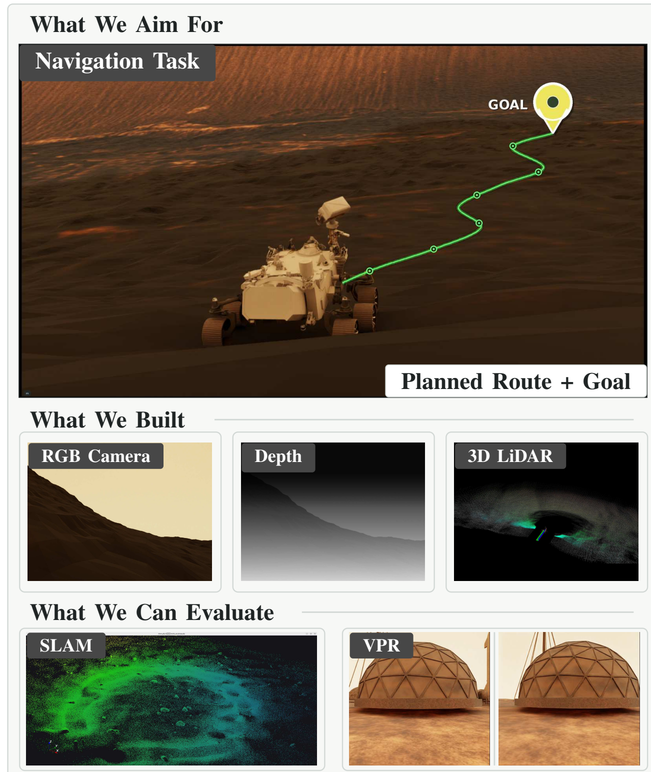


Fig. 1: **MarsLab overview.** MarsLab supports Mars-rover autonomy and navigation algorithm development through HiRISE-based customizable scenes and environment variables, together with sensor topic streams. Users can vary terrain, rocks, craters, solar illumination, and atmospheric dust (Fig. 4–Fig. 6), subscribe to RGB, depth/RGB-D, and 3D LiDAR sensor topics from the rover runtime, and evaluate autonomy modules including SLAM and visual place recognition, as demonstrated in our experiments.

Because these developments cannot be reproduced through field testing on Mars, simulation-based validation is essential for evaluating autonomy stacks before deployment. However, as summarized in Table I, existing resources rarely combine openness with full-stack autonomy and navigation support: mission simulators are typically not released as community tools [9], [10], several prior environments emphasize narrower task settings such as SLAM, photorealism, Reinforcement Learning (RL), or synthetic data [11]–[14], and analog datasets provide fixed recordings [15].

To address this gap, we present **MarsLab**: an open-source, ROS2 [16]-native Mars rover simulator for repeatable auton-

TABLE I
Concise survey of Mars rover simulation platforms and benchmark support.
Column abbreviations and symbols: **gen.** = generation; **seqs.** = sequences; **sens.** = sensor modalities; ✓ = reported/core capability; ○ = partial, indirect, or not primary; ✗ = not reported; **D** = depth; **L** = 3D LiDAR; **seg.** = semantic or instance labels; **NR** = not reported.

Simulator	Year	Scene gen.	Engine	Goal	Scenes/Seqs.	Rock	Dust	Light	Sens.	ROS2	Open
ROAMS [9]	1999 / 2004	Mars-like terrain & mission simulation	Custom / JPL DARTS-DSHELL	Rover mobility, dynamics, onboard SW V&V	Mission/testbed scenarios; Monte Carlo possible	✓	✗	○	Sensor/actuator subsystems; arm, power, nav/ locomotion SW	✗	○
ENav [10]	2020	Mars 2020 ENav test terrains	ROS-based + HDSim/RSVP	Autonomous nav V&V	Mars 2020 test scenarios / Monte Carlo runs	○	✗	○	Stereo/NavCam point cloud; camera rendering	✗	✗
Giubilato et al. [11]	2020	Simulated Martian / planetary-like environment	ROS/Gazebo	Visual & LiDAR SLAM evaluation	Long/Short simulated rover sequences / datasets	✓	✗	○	Cam / 3D L / IMU; stereo + LiDAR seqs.	✗	✓
MarsSim [12]	2023	Multiscale Mars terrain simulation	ROS/Gazebo	High-fidelity physical & visual rover simulation	NR	✓	○	✓	RGB/visual; locomotion data; D/L unclear	✗	✗
ISMRS [13]	2024	Digital-twin and asset-based scenes	Isaac Sim	Mars rover simulation + ML data synthesis	procedural generated scenes; rosbags; seqs. NR	✓	✗	✓	RGB/stereo/ point cloud; LiDAR/ranger, seg.; VSLAM demo	✓	✗
RLRoverLab [14]	2024	Synthetic rover training scenes	Isaac Sim / Omniverse + ORBIT; current repo Isaac Lab	RL navigation / manipulation / control	RL task scenes	○	✗	○	Task-dependent; RGB-D cam, height scan, Isaac obs.	✗	✓
MarsLab (Ours)	**2026**	**HiRISE-based + customizable**	**Isaac Sim ROS2**	**Rover autonomy algorithm development**	**Customizable multi-scenes; SLAM/VPR/nav tests**	✓	✓	✓	**RGB/D/L; wheel odom; GT logs**	✓	✓

omy evaluation across customizable terrain, lighting, dust, sensing, rover motion, and time-aligned ground truth pose data. MarsLab supports SLAM and navigation algorithm development through HiRISE-based customizable scenes [17] and standard rover sensor topics (Fig. 1).

Our contributions are:

1) **MarsLab**, an open-source, ROS2-native Mars rover simulation and validation environment built on NVIDIA Isaac Sim;
2) a scientifically grounded scene-generation pipeline that couples HiRISE-derived terrain models with procedural Mars-like rocks, craters, illumination, and atmospheric dust to produce Mars-relevant environmental variation;
3) a reproducible, multi-scene benchmark for SLAM, localization, place recognition, planning, and navigation, demonstrated through SLAM and VPR experiments.

II. Related Work

A. Planetary Simulation and Autonomy Validation

Simulation is an established approach for planetary-surface validation, but existing Mars tools remain limited in scope and availability. ROAMS and the Mars 2020 ENav simulator are mission-oriented and not community benchmarks [9], [10]. ROS/Gazebo and Isaac Sim systems such as Giubilato et al. [11], MarsSim [12], ISMRS [13], and RLRoverLab [14] address SLAM evaluation, visual/physical fidelity, digital twins, or RL, but do not jointly provide an open ROS2 benchmark with customizable Mars terrain, illumination, dust, sensors, and ground truth poses. Table I positions MarsLab against this landscape.

B. Planetary Datasets, Semantics, and Traversability

Planetary datasets capture important perceptual information, but their recorded trajectories, environmental conditions, and sensor configurations are typically fixed after release. AI4MARS supplies terrain labels for Mars imagery [19], while analog datasets such as the Canadian energy-aware navigation dataset and DLR S3LI add rover motion and multi-sensor logs [15], [20]. They support offline perception studies, but cannot vary GT pose trajectories, terrain statistics, dust, illumination, or sensor suites to measure closed-loop effects.

C. Long-Term Localization and Mars SLAM

Long-term localization is central to planetary autonomy because drift propagates to mapping, traversability assessment, and planning. Mars surface systems have used visual odometry [5], orbital-map matching for global correction [7], and onboard localization on real Mars data [6], but direct validation on Mars remains limited by cost, sparse ground truth pose data, and non-repeatable conditions.

MarsLab addresses this gap with HiRISE-derived and procedural terrain, a Perseverance-class rover, controlled illumination and dust, synchronized ground truth poses, and a standardized ROS2 benchmark.

III. MarsLab Architecture

A. System Overview

MarsLab is an open-source, ROS2-native simulator built on NVIDIA Isaac Sim 5.1 [21]. As shown in Fig. 2, the pipeline starts from reusable terrain, surface, and rover assets; constructs customizable Mars scenes; runs them with Isaac Sim physics and ROS2 streams; and exports experiment data for evaluation. Each run is specified by a Universal

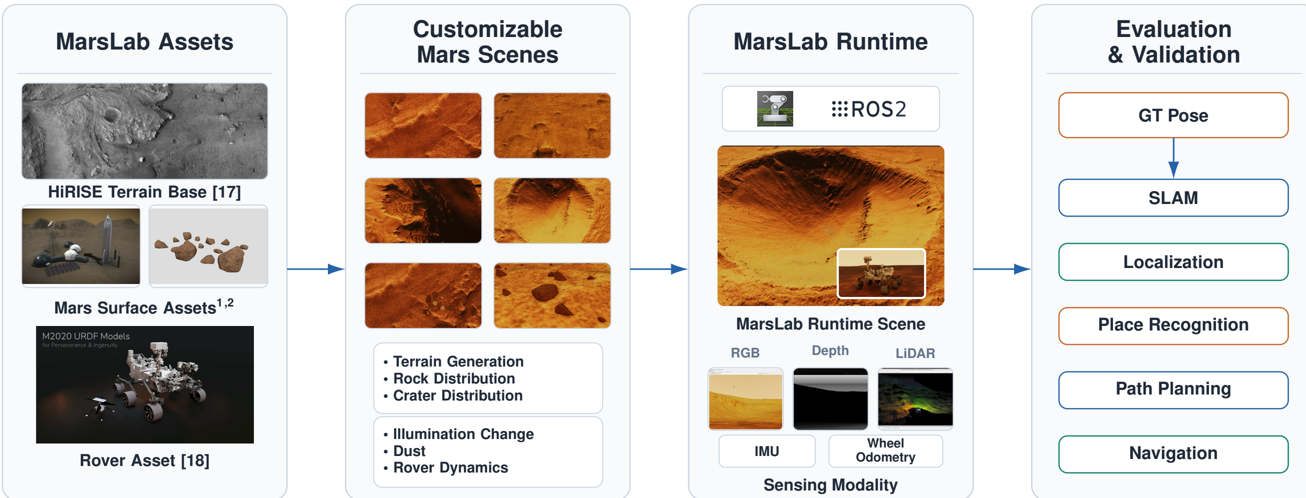


Fig. 2: **MarsLab pipeline.** MarsLab converts planetary terrain, surface assets, and rover models into customizable Mars experiments. The pipeline builds scene variants with controlled terrain generation, rocks, craters, illumination, and dust; executes the rover in an Isaac Sim/ROS2 runtime under Mars gravity; and exports sensor, command, and GT pose for SLAM, localization, place recognition, planning, and navigation evaluation.

Scene Description (USD) scene, YAML runtime configuration, seed, and trajectory/ground-truth pose data. Runtime parameters are validated before launch, and randomized stages such as crater sampling, rock placement, and sensor noise share the explicit seed for reproducible replay.

The architecture is organized as an end-to-end autonomy-evaluation pipeline. The scene-construction layer builds Mars-relevant worlds from terrain sources and customizable surface and environmental parameters. The rover-runtime layer loads each world into Isaac Sim, applies Mars gravity, and provides rover dynamics, command topics, sensor streams, and coordinate frames. The benchmark layer exports GT trajectory and sensor data for evaluating SLAM, localization, navigation, VPR, and related autonomy algorithms. This separation lets users vary scene conditions without changing the rover interface or the evaluation format.

B. Terrain and Scene Generation

MarsLab combines real and procedural terrain sources. HiRISE Digital Elevation Model (DEM)s and co-registered orthomosaics provide real terrain geometry and texture information, with Mastcam-Z imagery as a surface-appearance reference [17], [22]. Procedural terrains are synthesized with `synthterrain` [23] and augmented with the rock and crater models below. The reference scenes span plains, craters, canyons, and habitat sites[1] (Fig. 3; Table II), while Fig. 4–6 show the customizable terrain, illumination, and dust factors used as controlled evaluation conditions.

Scene construction follows a deterministic asset-build process. Given a crop region, terrain scale, and generation seed, the builder places rocks, crater fields, rover assets, lighting, and atmospheric settings into a single world description. The generated scene is stored separately from the runtime configuration, so it can be replayed with different sensors or controllers and the same GT trajectory can be evaluated under modified lighting or dust. This design keeps MarsLab from being a fixed dataset: it is a controlled generator of related experiments whose differences are known before evaluation.

C. Rock Distribution

Rock populations follow the Golombek–Huertas cumulative fractional-area model [24]:

$$F_k(D) = k\, e^{-q(k)\, D}, \qquad q(k) = 1.79 + \frac{0.152}{k}, \quad (1)$$

where D is rock diameter, k is the total rock-covered area fraction, and $F_k(D)$ is the cumulative area fraction covered by rocks with diameter at least D. Sampled diameters are used to scale USD rock assets[2] on the DEM surface, so the generated rock field follows the target size–frequency statistics rather than arbitrary placement (Fig. 4a).

D. Crater Distribution

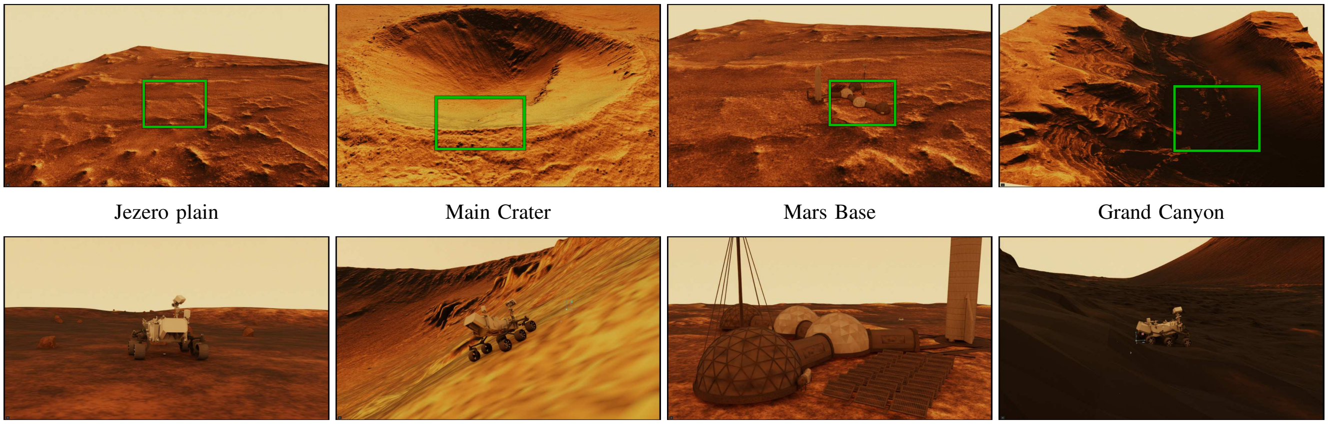
Crater diameters are sampled from the Hartmann–Neukum Mars production function [25]. For the valid $D \geq 10\,\mathrm{m}$ regime, MarsLab converts the cumulative SFD $C(D)$ into a normalized sampling CDF:

$$F(D) = 1 - \frac{C(D)}{C(D_{\min})}, \qquad D \in [D_{\min}, D_{\max}]. \quad (2)$$

Optional resurfacing masks follow crater-population clustering analysis [26], and diameter-dependent profiles use published fresh-crater morphometry [27]; thus crater density and shape remain tied to planetary chronology models (Fig. 4b).

[1] "Mars Base" by MOJackal, via Sketchfab, `https://sketchfab.com/3d-models/mars-base-2bc6660c63b24158bdded20ff5d2235e`, licensed under CC BY 4.0.

[2] "Mars Rocks" by Ivan Vakulko (milos4), via Sketchfab, `https://sketchfab.com/3d-models/mars-rocks-9f5c946255a24f1cb630ea95dceea587`, licensed under CC BY 4.0.



Jezero plain | Main Crater | Mars Base | Grand Canyon

Jezero rim | Crater interior | Base assets | Canyon wall

(a) | (b) | (c) | (d)

Fig. 3: **Reference MarsLab scenes.** MarsLab can generate additional terrain instances, and these four reference scene pairs are provided as scenes for evaluation. The green rectangle in each upper image indicates the approximate area represented by the corresponding lower scene: (a) a plain scene from Jezero crater, (b) a main-crater scene from Arcadia Planitia, (c) a Mars Base scene that augments the Jezero terrain with a habitat asset, and (d) a Grand Canyon scene from Elysium Planitia. All scenes are built on HiRISE-derived terrain, providing diverse geometry, texture, and structure for SLAM and place recognition stress testing.

TABLE II

MarsLab reference scenes. Compact summary of the four evaluation scenes in Fig. 3. Rock distribution, illumination, and atmospheric dust denote user-customizable scene parameters, including rock size–frequency/spatial-density settings, solar azimuth/elevation, and dust optical depth.

Fig.	Reference scene	Scene size	Rock	Dust	Illumination	Target Use
(a)	**Jezero plain/rim**	$0.5{\times}0.5\,\mathrm{km}$	✓	✓	✓	Navigation, terrain following, and low-texture localization baselines.
(b)	**Main Crater**	$3.6{\times}3.6\,\mathrm{km}$	✓	✓	✓	Repetitive-terrain failure analysis and rocky slope traversal.
(c)	**Mars Base**	$1.0{\times}1.0\,\mathrm{km}$	✓	✓	✓	Landmark- and feature-rich localization and VPR benchmarking.
(d)	**Grand Canyon**	$3.0{\times}3.0\,\mathrm{km}$	✓	✓	✓	Long-range drift evaluation over extended canyon terrain with strong elevation changes and complex geometry.

E. Solar Illumination and Dust Environment

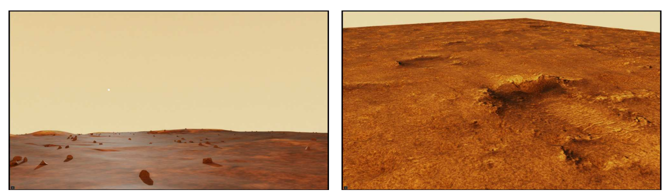
Illumination is treated as a customizable experimental variable. Direct surface irradiance follows Beer's law with Mars dust optical depth τ:

$$I(z) = \begin{cases} I_0\, e^{-\tau\, m(z)}, & z < 90^\circ, \\ 0, & z \geq 90^\circ, \end{cases} \tag{3}$$

where z is solar zenith angle, $I_0 \approx 589\,\mathrm{W\,m^{-2}}$, and $m(z)$ is the Appelbaum–Flood relative optical air mass [28]. Diffuse sky illumination follows a τ-indexed COMIMART reduction [29]. The same trajectory can then be replayed with customizable solar azimuth/elevation and dust optical depth: sun changes redistribute shadows (Fig. 5), while increasing τ reduces contrast and visibility (Fig. 6) without changing geometry. By varying one environmental parameter at a time, such as sun position or dust optical depth, MarsLab enables vision-based algorithms to be tested under diverse conditions within the same scene.

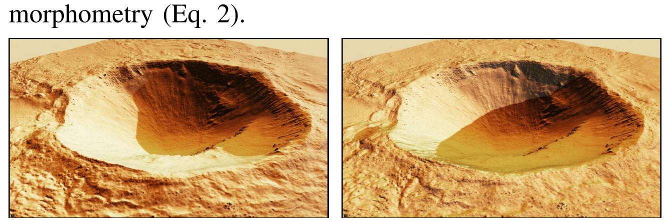
F. Rover Runtime and Physics

The runtime uses a Perseverance (M2020) model from NASA-JPL's public URDF release [18], converted to USD and simulated under Mars gravity ($3.72\,\mathrm{m/s^2}$). A sensor



(a) Rocky scene. (b) Crater scene.

Fig. 4: **Procedurally generated rock and crater scenes.** Rendered MarsLab terrain showing (a) the rock size–frequency distribution (Eq. 1) and (b) crater density and morphometry (Eq. 2).

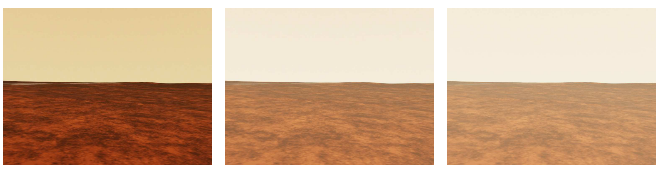


(a) Azimuth 70°, elevation 13° (b) Azimuth 360°, elevation 13°

Fig. 5: **Controllable solar illumination.** The sun azimuth and elevation are explicit experimental variables; changing the solar position redistributes cast shadows and shading across the same terrain, directly affecting feature visibility for vision-based navigation algorithms.

TABLE III
Customizable MarsLab scene generation. Beyond the reference scenes, MarsLab lets users build HiRISE-based scenes by controlling terrain extent, surface clutter, crater fields, illumination, and atmospheric dust. The table summarizes the scene-generation factors exposed as user-controlled variables.

Stage	**User-provided input**	**Customizable variation**
Terrain base	HiRISE DEM/orthomosaic	Crop region, terrain scale, and generation seed.
Surface clutter	Rock assets and rock size–frequency parameters	Rock density, diameter range, spatial sampling, and placement seed.
Crater field	Mars crater production-function parameters	Crater density, diameter range, morphology, and resurfacing masks.
Illumination	Solar-position and irradiance configuration	Sun azimuth, elevation, shadow direction, and scene shading.
Atmosphere	Mars dust optical-depth setting	Dust optical depth τ, sky brightness, and image contrast.



(a) $\tau = 0.3$ (b) $\tau = 3.0$ (c) $\tau = 6.0$

Fig. 6: **Atmospheric-dust sweep.** MarsLab uses the optical-depth parameter τ in Eq. 3 as a controlled evaluation condition for vision-based algorithms: increasing τ brightens the sky and ground, lowers image contrast, and reduces visibility while leaving the terrain geometry unchanged.

graph and `rclpy` bridge subscribe to `cmd_vel` and publish odometry, IMU, RGB, depth, RGB-D point cloud, 2D/3D LiDAR, TF, robot description, and ground-truth poses in REP-103/REP-105 frames (Fig. 7). GT trajectories are written in TUM format and replayed by a pure-pursuit follower with bounded commands ($v \leq 0.5\,\mathrm{m/s}$, $|\omega| \leq 0.2\,\mathrm{rad/s}$, $50\,\mathrm{Hz}$), giving a reference data stream for Absolute Trajectory Error (ATE)/RPE evaluation. The resulting sensor streams and GT trajectory data provide the common interface used by the runtime and the benchmarks below.

Overall, MarsLab connects customizable Mars scene construction, rover runtime simulation, and benchmark data export into a single autonomy-evaluation pipeline. The scene layer controls terrain, rocks, craters, illumination, and dust; the runtime layer provides standard rover command and

```
marslab@ros2:~$ ros2 topic list
/rover/cmd_vel[subscribe]                 /rover/imu[publish]
/clock[publish]                           /rover/imu_noisy[publish]
/tf[publish]                              /rover/rgb/image_raw[publish]
/tf_static[publish]                       /rover/rgb/camera_info[publish]
/rover/joint_states[publish]              /rover/depth/image_raw[publish]
/rover/robot_description[publish]         /rover/depth/points[publish]
/rover/odom[publish]                      /rover/lidar/points[publish]
/rover/GT_Trajectory[publish]             /rover/scan[publish]
```

Fig. 7: **MarsLab ROS2 topic interface.** Default subscribed and published topics exposed by the rover runtime.

TABLE IV
ATE RMSE (m) across MarsLab scenes and SLAM stacks. Camera methods are run at low (τ=0.5) and high (τ=6.0) dust; LiDAR is run once per scene. Per-row best in **bold**; ✗ marks a tracking failure.

	ORB-SLAM		RTAB-Map		MOLA
Scene	τ=0.5	τ=6.0	τ=0.5	τ=6.0	LiDAR
Mars Base	0.59	4.70	0.48	0.64	**0.20**
Main Crater Interior	40.28	✗	8.84	10.61	**0.13**
Grand Canyon	✗	✗	12.28	19.08	**0.38**

✗ = monocular tracking failure (no usable trajectory).

sensor topics; and the benchmark layer records GT trajectory data for evaluating SLAM, localization, navigation, VPR, and related autonomy algorithms. This structure lets users vary environmental conditions while keeping the rover interface and evaluation data consistent across experiments.

IV. Experiments

We evaluate MarsLab with two studies: a multi-scene SLAM benchmark comparing RGB, RGB-D/wheel-odometry, and 3D-LiDAR methods under controlled dust, and a VPR benchmark over repeated *Mars Base* traversals.

A. SLAM Benchmark

Setup. The SLAM benchmark uses three representative scenes: *Mars Base* for flat terrain with structural landmarks, *Main Crater Interior* for rock-scattered crater terrain, and *Grand Canyon* for large elevation changes and occlusion. We evaluate ORB-SLAM [30] on monocular RGB, RTAB-Map [31] on RGB-D plus wheel odometry, and MOLA [32] on 3D LiDAR. ORB-SLAM and RTAB-Map are run at τ=0.5 and τ=6.0; MOLA is run once per scene because the dust model affects radiance but not range.

Protocol. For each scene, we design the GT trajectory to keep the rover on traversable terrain by accounting for rock distribution and local terrain slope. The pure-pursuit follower executes the trajectory, and MarsLab logs ground truth poses in TUM format. We report ATE RMSE after Umeyama SE(3) alignment. Table V and Fig. 10 define the GT trajectories, Fig. 8 overlays trajectories, and Fig. 9 visualizes reconstructed maps. All runs use Isaac Sim 5.1, ROS2 Jazzy, and an NVIDIA RTX 5070 Ti workstation.

Results. Table IV shows a clear modality ordering. MOLA remains sub-metre on all scenes (0.13–$0.38\,\mathrm{m}$), RTAB-Map ranges from $0.48\,\mathrm{m}$ on *Mars Base* to 8–$19\,\mathrm{m}$ on rugged scenes, and monocular ORB-SLAM is fragile, reaching $40.3\,\mathrm{m}$ on *Main Crater Interior* and losing tracking on *Grand Canyon*. Increasing dust from τ=0.5 to τ=6.0 mainly affects vision: ORB-SLAM grows $7.9\times$ on *Mars Base* and fails on *Main Crater Interior*, while RTAB-Map degrades more gradually because depth and wheel odometry anchor scale and motion. MOLA is unchanged, consistent with a dust model that attenuates radiance but not range.

Failure modes. Scene geometry also matters. *Mars Base* is easiest for vision, whereas canyon walls and self-similar crater terrain starve monocular features, causing the tracking losses marked in Fig. 8 and Table IV. MOLA's error instead

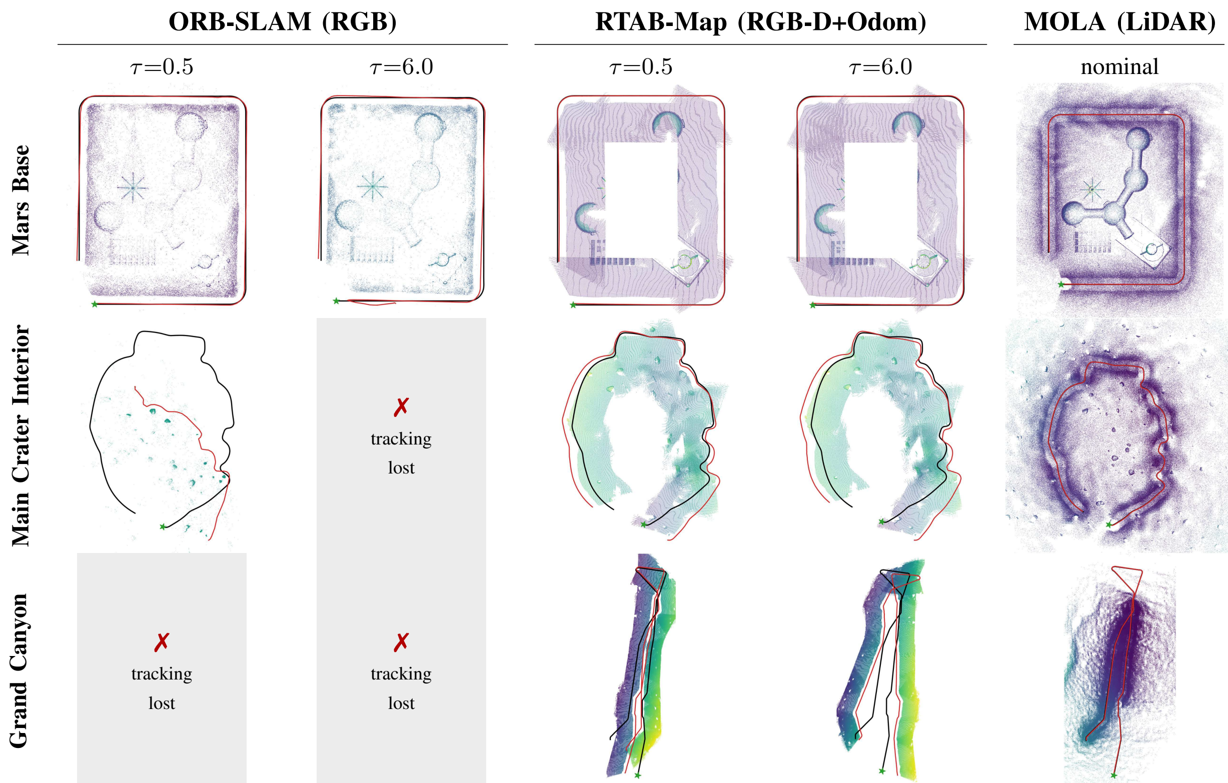


Fig. 8: **Multi-scene SLAM benchmark: estimated vs. ground-truth trajectories.** Top-down view of every run—the estimated trajectory (red) versus ground truth (black) after Umeyama alignment, over the reconstructed point-cloud map (coloured by height); rows are scenes, columns are sensing modality and dust optical depth. Grey tiles (✗) mark monocular ORB-SLAM runs that lost tracking and produced no usable trajectory. Scene names and GT trajectory lengths are listed in Table V; GT trajectory geometry is shown in Fig. 10; per-run ATE in Table IV; reconstructed 3D maps in Fig. 9.

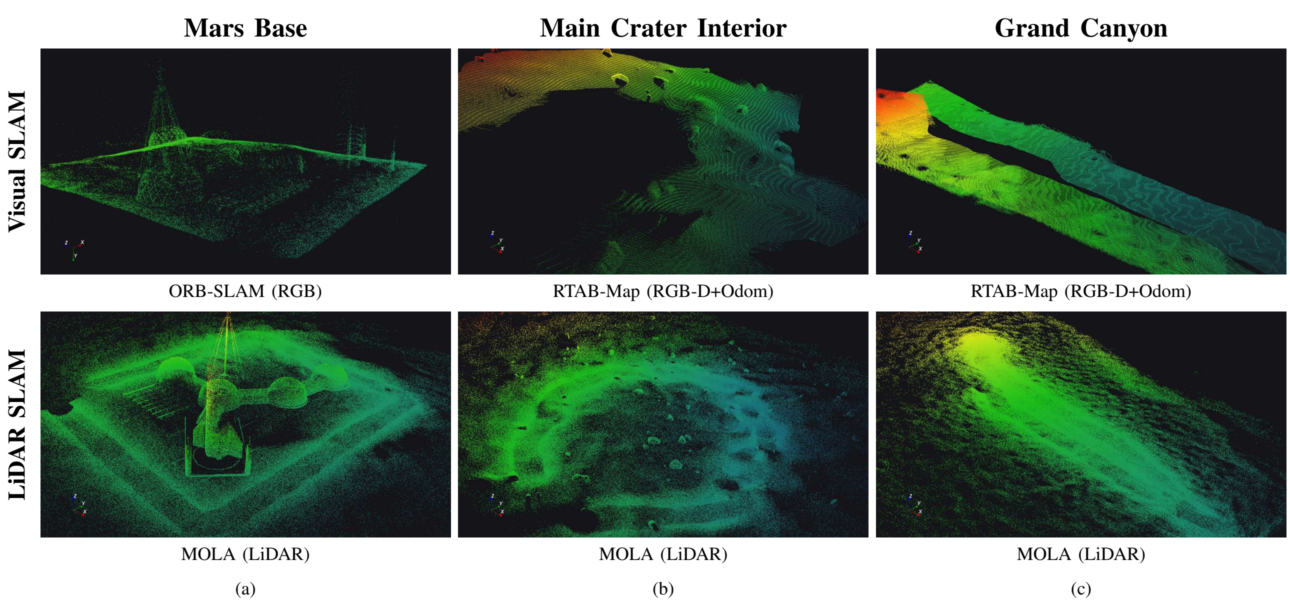
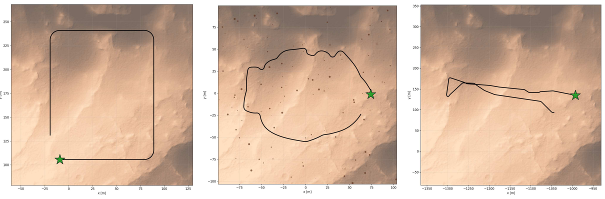


Fig. 9: **Reconstructed 3D maps: Visual SLAM vs. LiDAR SLAM.** Point-cloud maps reconstructed for each scene, comparing visual SLAM (top) against LiDAR SLAM (bottom; MOLA). For (a) *Mars Base* the visual-SLAM map is from ORB-SLAM (RGB); for (b) *Main Crater Interior* and (c) *Grand Canyon* it is from RTAB-Map (RGB-D+Odom), whose depth and wheel-odometry inputs remained stable where monocular ORB-SLAM lost tracking. Points are coloured by elevation.



Mars Base Main Crater Grand Canyon

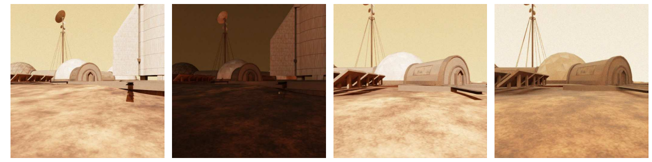
—— route ★ start point

Fig. 10: **GT trajectories.** Compact overview of the three indexed ground-truth trajectories used in the SLAM benchmark.

TABLE V
SLAM benchmark GT trajectory metadata. Each GT trajectory is authored in a MarsLab scene, replayed by the rover runtime, and logged from Isaac Sim pose streams in TUM format for ATE/RPE evaluation.

Scene	Length	GT Trajectory Role
Mars Base	$\approx 440\,\mathrm{m}$	Feature- and landmark-rich trajectory for testing localization in structured habitat scenes.
Main Crater Interior	$\approx 393\,\mathrm{m}$	Rock-rich trajectory for testing localization under repetitive terrain and sparse visual features.
Grand Canyon	$\approx 648\,\mathrm{m}$	Long-distance trajectory for testing drift across extended canyon terrain with large elevation changes.

Query Retrieved (R@1) Query Retrieved (R@1)



(i) Day/Dark, τ=0.05 (ii) Dust, τ=0.05 to τ=0.60

Fig. 11: **Visual place recognition: query vs. retrieved.** Qualitative examples for the two Table VI protocols: (i) day-lit queries matched to a dark database at τ=0.05 and (ii) low-dust daytime queries at τ=0.05 matched to high-dust daytime database images at τ=0.60. Each pair shows the query frame and the top-1 retrieved database frame (R@1).

TABLE VI
Visual place recognition recall on the MarsLab Mars-base benchmark using the strict $2\,\mathrm{m}$ positive radius. Results are Recall@N (%). Best values in each protocol/metric are shown in bold.

Method	Day/Dark, τ=0.05			Dust, τ=0.05→0.60		
	R@1	R@5	R@10	R@1	R@5	R@10
NetVLAD [33]	77.90	95.35	98.36	85.02	98.18	99.33
AnyLoc [34]	64.04	92.21	97.17	74.89	95.71	99.06
BoQ [35]	**91.44**	**99.19**	**99.80**	**94.65**	**99.37**	**99.78**

follows GT trajectory length, rising from $0.13\,\mathrm{m}$ on the $393\,\mathrm{m}$ *Main Crater Interior* trajectory to $0.38\,\mathrm{m}$ on the $648\,\mathrm{m}$ *Grand Canyon* trajectory, or only 0.03–0.06% of distance travelled. This separation of dust, geometry, and GT trajectory length illustrates the controlled attribution MarsLab is designed to support.

B. Visual Place Recognition

Setup. Beyond trajectory-level SLAM, we evaluate VPR as image retrieval over repeated *Mars Base* traversals. We compare NetVLAD [33], AnyLoc [34], and BoQ [35] on 4456 query and 4455 database frames recorded every $0.1\,\mathrm{m}$. A retrieval is correct when one of the top-N database images lies within $2\,\mathrm{m}$ of the query pose; we report Recall@N for $N \in \{1, 5, 10\}$.

Protocol. We test two appearance changes on the same Mars Base route. *Day/Dark* matches day-lit queries to a dark database at τ=0.05, while *Dust* matches low-dust daytime queries at τ=0.05 to high-dust daytime database images at τ=0.60.

Results. Table VI shows that BoQ performs best in both protocols, reaching 91.44% R@1 for Day/Dark and 94.65% R@1 for Dust. NetVLAD ranks second across all metrics, and AnyLoc recovers by R@10 despite lower R@1 accuracy. Fig. 11 confirms the same trend qualitatively: the top-1 matches for the habitat dome and lander/Starship still depict the correct places under viewpoint and appearance changes.

C. Experiment Summary

Taken together, the experiments show how MarsLab supports autonomy benchmarks across both trajectory-level estimation and image-level place recognition. The SLAM benchmark varies dust level, long-range GT trajectory, rock distribution, and scene structure, spanning feature- and landmark-poor terrain as well as landmark-rich *Mars Base* conditions. It also exposes the sensing modalities needed for practical rover navigation: monocular camera, RGB-D, LiDAR, and wheel odometry, with IMU streams available for additional tests even though this paper does not include an IMU-based SLAM run. The VPR benchmark complements this by using the same *Mars Base* route to change dust and illumination while preserving pose labels, allowing users to design controlled VPR protocols and compare retrieval behavior across methods. The different R@1 scores in Table VI confirm that these appearance changes produce meaningful algorithm-dependent differences rather than trivial retrieval cases.

V. Discussion

MarsLab focuses on reproducible Mars autonomy experiments with customizable environmental variation. Each trial specifies the scene, rover runtime, sensor topics, GT trajectory, and generation parameters, so users can change one factor at a time while keeping the remaining conditions explicit. This makes the benchmark useful beyond rendered images: SLAM experiments can connect drift and tracking failure to terrain geometry, dust, landmarks, and sensing modality, while VPR experiments can connect recall changes to illumination and atmospheric dust. These results show that MarsLab can support both trajectory-level and image-level autonomy evaluation within the same simulator.

Future work will extend MarsLab from a rover-centered benchmark into a broader Mars autonomy development platform. We will add heterogeneous robot support, including multiple rover types and aerial scouts, so users can evaluate systems that combine ground mobility with complementary viewpoints. We will also expand the task coverage beyond SLAM and VPR to support computer-vision tasks, traversability estimation, path planning, navigation-stack integration, and multi-robot cooperation. These extensions will help users develop and compare larger autonomy stacks while still using the same customizable Mars scenes, sensor interfaces, and GT trajectory data.

VI. Conclusion

We presented MarsLab, an open-source, ROS2-native Mars rover simulator for reproducible autonomy evaluation. MarsLab combines HiRISE-derived and procedural terrain with distributed rock, crater, illumination, and dust models, then plays each scene and GT trajectory through an Isaac Sim/ROS2 runtime with ground truth pose data. The experiments show that MarsLab lets users vary sensing modality, dust, scene geometry, and GT trajectory length while keeping the remaining evaluation conditions explicit. By coupling Mars-relevant scene generation with rover motion and standard robotics interfaces, MarsLab provides a practical testbed for localization, mapping, navigation, place recognition, and broader rover-autonomy algorithm development.

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